

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

June 2025 Semester End Main Examinations

Programme: B.E.

Branch: CS Cluster Except AIML and CSBS

Course Code: 23MA4BSLAO

Course: Linear Algebra and Optimization

Semester: IV

Duration: 3 hrs.

Max Marks: 100

- Instructions**
1. Each unit has an internal choice. Answer one complete question from each unit.
 2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - 1	CO	PO	Marks
	1	a)	Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$ and $v_3 = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$. Find an implicit description of hyperplane H that passes through v_1 , v_2 , and v_3 .	1	1	6
		b)	Obtain the critical points of $f(x, y, z) = xy + yz + zx - 4x + 2y$ and hence find the nature of the critical points of f using the Hessian matrix.	1	1	7
		c)	Given $f = \begin{bmatrix} x_0^2 x_1 x_2 & x_1^2 x_2 x_3 \\ x_3^2 x_1 x_2 & x_2^3 x_1 \end{bmatrix}$ and $X = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$, then verify the identity $\frac{\partial(\text{Trace}(f(X)))}{\partial X} = \text{Trace}\left(\frac{\partial f(X)}{\partial X}\right)$.	1	1	7
			OR			
	2	a)	Show that the set $S = \{(x, y) 2x + 3y = 5\} \subset R^2$ is a convex set.	1	1	6
		b)	Find the value of 'a' for which $P = (0,0,0)$ is the critical point of the function $f(x, y, z) = 2x^2 + 3y^2 + 4z^2 + axyz$ and hence find the nature of the critical point P using the Hessian matrix.	1	1	7
		c)	If $f = \begin{bmatrix} \sin(x_0 + 2x_1) & 2x_1 + x_3 \\ 2x_0 + x_2 & \cos(2x_2 + x_3) \end{bmatrix}$ and $X = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$ then verify the identity $\frac{\partial(f(X)^T)}{\partial X} = \left(\frac{\partial f(X)}{\partial X}\right)^T$.	1	1	7
			UNIT - 2			
	3	a)	Minimize $f(x) = x^2 - 2.6x + 2$, $x \in [-2, 3]$ using Fibonacci search method. Perform six iterations.	1	1	6
		b)	Find the maximum and minimum values of $f(x, y, z) = 4y - 2z$ subjected to the constraints $2x - y - z = 2$ and $x^2 + y^2 = 1$ using method of Lagrange multipliers.	1	1	7
		c)	Maximize $Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$ subjected to the constraints $x_1 + x_2 \leq 10$, $x_2 \leq 8$, $x_1, x_2 \geq 0$ by deriving the KKT conditions.	1	1	7
			OR			

4	a)	Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ using Newton-Raphson method by taking initial point as $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.	1	1	6														
	b)	Find the maximum and minimum values of $f(x, y, z) = x + y + z^2$ subjected to the constraints $x + y + z = 1$ and $x^2 + z^2 = 1$ using the method of Lagrange multipliers.	1	1	7														
	c)	Minimize $Z = (x - 3)^2 + (y - 2)^2$ subjected to the constraints $x^2 + y^2 \leq 5$, $x + 2y \leq 4$ and $x, y \geq 0$ by deriving the KKT conditions.	1	1	7														
		UNIT - 3																	
5	a)	Let W be a subspace of $\mathbb{R}^5$ spanned by $u = (1, 2, 3, -1, 2)$ and $v = (2, 4, 7, 2, -1)$. Find the basis of $W^\perp$.	1	1	6														
	b)	Find an orthogonal matrix P whose first row is $u_1 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$.	1	1	7														
	c)	Solve the system of equations $AX = B$ where $A = \begin{bmatrix} 2 & 3 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$ by QR factorization method.	1	1	7														
		OR																	
6	a)	Consider $f(t) = 3t - 5$, $g(t) = t^2$ in $P(t)$ and the inner product is defined as $\langle p, q \rangle = \int_0^1 p(t)q(t)dt$. Find $\langle f, g \rangle$, $\ f\ $ and $\ g\ $.	1	1	6														
	b)	Find the projection of $v = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ along $w = \begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix}$ in $M_{2 \times 2}$ with respect to the inner product $\langle A, B \rangle = \text{tr}(B^T A)$.	1	1	7														
	c)	A sales organization obtains the following data relating the number of salespersons to annual sales. <table border="1"><tr><td>Number of salespersons</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Annual Sales (millions of dollars)</td><td>2.3</td><td>3.2</td><td>4.1</td><td>5.0</td><td>6.1</td><td>7.2</td></tr></table> Let x denote the number of salespersons and let y denote the annual sales. Find the least squares line of the form $y = a + bx$ and hence estimate the annual sales when there are 14 salespersons.	Number of salespersons	5	6	7	8	9	10	Annual Sales (millions of dollars)	2.3	3.2	4.1	5.0	6.1	7.2	1	1	7
Number of salespersons	5	6	7	8	9	10													
Annual Sales (millions of dollars)	2.3	3.2	4.1	5.0	6.1	7.2													
		UNIT - 4																	
7	a)	Given $B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, find B^6 using Cayley-Hamilton theorem with minimal conventional operations.	1	1	6														
	b)	Find the algebraic and geometric multiplicities of each eigenvalue λ of the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x - y + z, x - y + z, x - 2y + 2z)$.	1	1	7														
	c)	Let $\Delta(t) = (t + 6)^6$ and $m(t) = (t + 6)^3$ be the characteristic and minimal polynomial of T respectively. Obtain all possible Jordan canonical forms of T .	1	1	7														
		OR																	

8	a)	Given $B = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$, find B^{-3} using Cayley-Hamilton theorem with minimal conventional operations.	1	1	6														
	b)	Find the eigenspace corresponding to each eigenvalue of $T: P_1 \rightarrow P_1$ defined by $T(at + b) = (3a + 5b)t - (2a + 4b)$.	1	1	7														
	c)	Find the characteristic and minimal polynomial of the matrix $\begin{bmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix}$.	1	1	7														
		UNIT - 5																	
9	a)	Obtain the transformation matrix that reduces the quadratic form $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ to its canonical form and hence find the rank and index of the quadratic form.	1	1	10														
	b)	Find the singular value decomposition of the matrix $\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.	1	1	10														
		OR																	
10	a)	Find the orthogonal modal matrix that diagonalizes the matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$.	1	1	10														
	b)	Apply principal component analysis to reduce the given 2-dimensional data to 1-dimensional data. <table border="1"><tr><td>x</td><td>1</td><td>5</td><td>2</td><td>6</td><td>7</td><td>3</td></tr><tr><td>y</td><td>3</td><td>11</td><td>6</td><td>8</td><td>15</td><td>11</td></tr></table>	x	1	5	2	6	7	3	y	3	11	6	8	15	11	1	1	10
x	1	5	2	6	7	3													
y	3	11	6	8	15	11													
